

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer. Possesses a Master of Science in Computer Science with expertise in building scalable microservices and distributed data pipelines across various domains. Demonstrates a strong command of Java, Python, Spark, SQL, AWS, and Azure, coupled with experience in CI/CD automation and graduate-level work on large-scale data platforms and privacy-preserving AI applications including LLMs and GDPR considerations. Eager to leverage this background to drive product development and technical strategy as a Founding Backend Engineer.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Built and automated CI/CD pipelines using Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments, adhering to product launch best practices.
- Modernized legacy systems by migrating to Log4j 2.x and improving monitoring and error handling, reducing issue resolution time by 20 percent.
- Led a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, ensuring robust data security and HIPAA compliance.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, contributing to product development initiatives.
- Improved system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, demonstrating a strong track record in efficient deployment.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering a collaborative environment.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage for critical application modules.
- Supported release planning and sprint execution using JIRA within Agile development teams, ensuring successful product launch coordination.

PROJECTS

NYU Taxi Data Analysis Platform January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze more than 8 GB of NYC taxi trip data in near real time.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization, showcasing Fastapi and UX considerations.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, with a focus on SaaS architecture.

Privacy-Preserving AI Email Client (Capstone Project) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, leveraging Large Language Models (LLMs).
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, ensuring SOC 2 compliance.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, considering data security best practices.

SKILLS

Programming Languages: Proficient in Java, Python, and SQL, with emerging experience in Go.

- Frameworks and Libraries: Expertise in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, and Fastapi.
- Web Services: Skilled in designing and implementing REST API and SOAP API solutions.
- Cloud Platforms: Experienced with AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and Google Cloud Platform, supporting SaaS models.
- Databases: Proficient in PostgreSQL, Oracle Database, IBM DB2, and RDS, with a strong understanding of database operations.
- Tools and Technologies: Adept with Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for infrastructure automation.
- Concepts: Strong grasp of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, UX, and Hardware integration.
- Data Analysis and Visualization: Skilled in Tableau and QuickSight for insightful data representation.
- Compliance and Security: Familiarity with GDPR, HIPAA, and SOC 2 standards, ensuring robust data security.